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Inventor(s)	Weiher; Keith L et al.

Radio frequency slab laser

Abstract

A radio frequency, RF, slab laser comprising a live electrode (**102**) and a ground electrode (**108**) whose inwardly facing surfaces face each other to form a gap for forming a plasma discharge when the live electrode is supplied with a suitable RF drive signal. The electrodes are enclosed in a vacuum space by a vacuum housing (**114**) with an access aperture (**116**). The access aperture is sealed with a vacuum flange (**70**) that comprises an electrically insulating connector. A plurality of hollow conductors (**62**) are arranged to extend through the vacuum flange into the vacuum space and connect with the live electrode. The hollow conductors connect to the live electrode to supply it with its RF drive signal and also coolant fluid which is distributed through fluid circulation channels (**80***a*, **80***b*). Coolant fluid is supplied to the live electrode through certain ones of the hollow conductors and taken out by others.

Inventors: **Weiher; Keith L (Wadena, MN), Kern; Gerald L (Wadena, MN), Colby; Jacob D (Wadena, MN), Jackson; Paul E (Wadena, MN)**

Applicant: **KERN TECHNOLOGIES, LLC** (Wadena, MN)

Family ID: **1000008781853**

Assignee: **KERN TECHNOLOGIES, LLC (Wadena, MN)**

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5123028	12/1991	Hobart	N/A	N/A
5164952	12/1991	Guenther	N/A	N/A
5596597	12/1996	Nishida	372/107	H01S 3/02
7558308	12/2008	Shackleton	N/A	N/A
8731015	12/2013	Newman	N/A	N/A
9263849	12/2015	Kern	N/A	N/A
2005/0175054	12/2004	Shackleton	372/87	H01S 3/0315
2005/0180483	12/2004	Taufenbach	372/87	H01S 3/038
2006/0029116	12/2005	Shackleton	372/55	H01S 3/09705
2010/0118901	12/2009	Newman	N/A	N/A
2010/0189156	12/2009	Clementi	N/A	N/A
2011/0182319	12/2010	Hua	372/55	H01S 3/134
2012/0033703	12/2011	Newman	N/A	N/A
2012/0106586	12/2011	Villarreal-Saucedo et al.	N/A	N/A
2015/0188282	12/2014	Kern	372/38.04	H01S 3/09702
2019/0181028	12/2018	Patel	N/A	N/A

Primary Examiner: Menefee; James A

Attorney, Agent or Firm: Nemphos Braue LLC

Background/Summary

FIELD OF THE INVENTION
(1) The present disclosure relates to a radio frequency (RF) slab laser.
BACKGROUND
(2) An RF slab laser is a laser with a kind of flat planar construction, which in its simplest form has only two electrodes. The two electrodes are arranged on top of each other so that their inwardly facing surfaces form a gap of a certain thickness. When an RF drive signal is applied to the

electrodes, a plasma discharge of a gas, such as carbon dioxide, is formed in the gap. The plasma provides a gain medium capable of supporting stimulated emission within a resonator cavity formed by end mirrors which are typically arranged just outside the gap. The electrodes, or at least the gap between them where the plasma is to be formed, needs to be in a vacuum. In most commercial designs, the plasma is induced between a “cold” ground electrode and an RF driven “live” or “hot” electrode. However, it is also possible to use two RF electrodes that are both “live” and driven with complementary phases.

(3) In RF slab lasers there is inevitably a heat management issue with the electrodes, in particular for the live electrode. For RF slab lasers with lower output powers, an internal conduction cooling path to an external air-cooled heat exchanger is generally adequate, whereas for higher laser output powers the electrodes typically need to be cooled by circulating a coolant fluid, typically water, around the electrode.

(4) Some known approaches for water cooling of high powered RF slab lasers are now summarized.

(5) U.S. Pat. No. 5,123,028 discloses a carbon dioxide slab laser with water cooling for the live electrode and the ground electrode. For each electrode, a coolant loop is provided by a copper pipe which is soldered into channels that have been machined out of the outwardly facing surfaces of the electrodes. The ends of the pipes pass through the end plate of the vacuum housing and are electrically grounded. The center segment of the coolant pipe that is in physical contact with the live electrode needs to be electrically isolated from the grounded part of the coolant pipe. This is done with a pair of in-line union connectors made of an electrically insulating material, so that the part of the pipe which is live is electrically isolated from the remainder of the pipe which is grounded. A drawback of the design of U.S. Pat. No. 5,123,028 is the need for the insulating union connectors. These add cost to the laser manufacture and are a potential water leak source as well as sometimes being an unwanted generator of RF discharge, since they form a junction between live and ground.

(6) U.S. Pat. No. 8,731,015B2 discloses a carbon dioxide slab laser which has a special design for cooling the live electrode involving a double-folded assembly of coolant tubes that are in thermal contact with the live electrode. The coolant tubes have a very high RF-impedance which avoids the need for them to be electrically insulated from the laser housing and so dispenses with the need for the insulating union connectors of U.S. Pat. No. 5,123,028. The coolant tube is dimensioned to extend beyond the end of the electrode through which it passes by an amount that provides a sufficiently high inductance at the RF frequency (e.g. about 100 MHz) that it can contact the grounded vacuum housing without causing a short circuit.

(7) US2010189156A1 discloses a carbon dioxide slab laser in which both the live and ground electrodes have their outwardly facing surfaces outside the vacuum enclosure. Coolant channels are formed in each electrode by four lengthways grooves formed in the outwardly facing surface. On each electrode an insert piece is screwed onto the outwardly facing surface to close the grooves in a liquid-tight manner and thereby form the coolant channels. For each electrode, the coolant fluid is introduced into the ends of the groove channels by a pair of connector pipes. The coolant fluid then flows along the electrode from one end to the other through the groove channels. The coolant fluid is then extracted from the other end by another pair of connector pipes.

(8) U.S. Pat. No. 9,263,849B2 discloses an alternative approach for designing a high power RF slab laser which can avoid the need for water cooling of the RF feed-through by improving the efficiency of the RF coupling into the live electrode and hence reducing the amount of heat that is generated in the RF feed-through. The outputs from four RF power amplifiers are passed through their own impedance matching circuits and then connected individually and separately to the live electrode. This design reduces the amount of heat generated by a factor ‘n’ compared with a design in which a single RF supply line is routed into the vacuum enclosure and contacted to the live electrode via a single impedance matching circuit, e.g. a factor of four in the example of having

four pairs of RF power amplifier and impedance matching circuit. This improvement is achieved because the heat generated by the impedance matching circuit is proportional to the square of the current, so having ‘n’ independent current injectors each carrying ‘I/n’ of the total current injected into the live electrode reduces the heating by a factor ‘n’ compared with having a single current injector supplying the live electrode with the whole current.

(9) U.S. Pat. No. 5,164,952 discloses an RF slab laser with a live electrode and a ground electrode arranged in a vacuum housing. Metal coolant tubes feed coolant fluid through an end wall of the laser into an RF-tight antechamber that is adjacent one end of the vacuum housing. The metal coolant tubes are joined to short electrically insulating sections in the antechamber whose other ends are joined to further metal tubes. Within the antechamber, the further metal tubes are connected to a T-shaped conductive clamp that transfers RF power to the further metal tubes. The further metal tubes are then fed through into the vacuum chamber via a vacuum-tight ceramic disc that insulates the vacuum housing from the RF. The further metal tubes once inside the vacuum chamber then bend through 90 degrees to connect onto the top surface of the live electrode, thereby supplying the live electrode with RF power as well as supplying (and removing) coolant fluid to (and from) the live electrode.

BRIEF SUMMARY OF THE INVENTION

(10) According to one aspect of the disclosure, there is provided an RF slab laser comprising: a first electrode and a second electrode having respective inwardly and outwardly facing surfaces, wherein their respective inwardly facing surfaces face each other and are spaced apart by a gap forming a slab waveguide of a thickness dimensioned to allow a plasma discharge to be formed by driving at least the first electrode with an RF drive signal, the first electrode being provided with at least one fluid circulation channel for distributing coolant fluid around the first electrode, a vacuum housing enclosing at least the inwardly facing surfaces of the first and second electrodes inside a vacuum space; and a plurality of hollow conductors connected to the outwardly facing surface of the first electrode to supply the RF drive signal to the first electrode and the coolant fluid to the at least one fluid circulation channel of the first electrode. Moreover, the vacuum housing has an access aperture adjacent the outwardly facing surface of the first electrode. Further, the vacuum flange is arranged in the access aperture to form a vacuum-tight seal with the vacuum housing and a further vacuum-tight seal with the outwardly facing surface of the first electrode.

(11) In one embodiment, the vacuum flange comprises a sleeve, an electrically insulating connector and a base, the base having an intermediate length portion of each hollow conductor embedded therein in a vacuum-tight manner, and the electrically insulating connector being connected with respective vacuum-tight connections between the sleeve and the base, wherein the sleeve forms said vacuum-tight seal with the vacuum housing and wherein the base forms said further vacuum-tight seal with the outwardly facing surface of the first electrode.

(12) In another embodiment, the vacuum flange comprises an electrically insulating connector extending from the access aperture, where it forms said vacuum-tight seal with the vacuum housing, to the outwardly facing surface of the first electrode, where it forms said further vacuum-tight seal with the outwardly facing surface of the first electrode, thereby defining a portion of the outwardly facing surface of the first electrode that is outside the vacuum space to which the hollow conductors are connected.

(13) In a further embodiment, the vacuum flange comprises a sleeve and an electrically insulating connector, wherein the sleeve forms said vacuum-tight seal with the vacuum housing and wherein the insulating connector forms said further vacuum-tight seal with the outwardly facing surface of the first electrode.

(14) In other embodiments, at least a portion of the outwardly facing surface of the first electrode is outside the vacuum space and is used to connect the hollow conductors. The hollow conductors thus remain outside the vacuum space obviating the need for a vacuum feed-through. This may be implemented by the vacuum housing being provided with an access aperture and by providing an

electrically insulating connector that extends from the access aperture, where it forms a vacuum-tight seal, to the outwardly facing surface of the first electrode, where it forms a further vacuum-tight seal. Said portion of the outwardly facing surface of the first electrode that is outside the vacuum space is thus formed by the access aperture in combination with the electrically insulating connector which may be hollow, e.g. a cylindrical tube that is dimensioned to match a circular-shaped access aperture.

(15) There are different alternatives for forming the fluid circulation channels. In some embodiments, each fluid circulation channel comprises at least one internal passage formed within the first electrode which terminates in apertures in a surface of the first electrode, for example the outwardly facing surface of the first electrode, for coolant fluid input and output, the fluid input and output apertures being arranged in fluid-flow connection with proximal ends of the hollow conductors. One implementation of such embodiments is for the fluid input and output apertures to be arranged aligned with the proximal ends of the hollow conductors. **1***s* In other embodiments, the fluid circulation channel(s) comprise further hollow conductors arranged in thermal contact with the outwardly facing surface of the first electrode (e.g. by soldering as described for the live electrode in the above-referenced U.S. Pat. No. 5,123,028) and in fluid-flow connection with the hollow conductors of the vacuum feed-through. The hollow conductors may be formed integrally with the further hollow conductors, i.e. as single hollow conductors, similar to what is shown for the live electrode in the above-referenced U.S. Pat. No. 5,123,028. In such embodiments, the coolant channels are thus formed by hollow conductors rather than internal coolant channels formed within the electrode(s).

(16) In some embodiments, the first electrode has an elongate shape with a length several times greater than its width (e.g. at least 5 or 10 times) and the hollow conductor connections are positioned at least approximately mid-way along the outwardly facing surface of the first electrode, thereby subdividing the first electrode into first and second arms. Each arm is preferably provided with at least one fluid circulation channel, namely a first fluid circulation channel extending around the first arm of the first electrode and a second fluid circulation channel extending around the second arm of the first electrode.

(17) In one particular implementation, the first fluid circulation channel is connected to be supplied with coolant fluid by first and second ones of the hollow conductors and the second fluid circulation channel may be connected to be supplied with coolant fluid by third and fourth ones of the hollow conductors. This is a 4-pin arrangement of hollow conductors which allows the first electrode to be supplied directly with four independent RF supplies, i.e. the RF drive signal has four independent components which are not combined but rather are connected independently to the first electrode by four of the hollow conductors. Here there may be first and second fluid circulation channels extending around the first arm of the first electrode and third and fourth fluid circulation channels extending around the second arm of the first electrode.

(18) In another particular implementation, the first fluid circulation channel is connected to be supplied with coolant fluid by first and second ones of the hollow conductors and wherein the second fluid circulation channel is connected to be supplied with coolant fluid by third and fourth ones of the hollow conductors, wherein the third fluid circulation channel is connected to be supplied with coolant fluid by fifth and sixth ones of the hollow conductors and wherein the fourth fluid circulation channel is connected to be supplied with coolant fluid by seventh and eighth ones of the hollow conductors. This is an 8-pin arrangement of hollow conductors which allows the first electrode to be supplied directly with eight independent RF supplies, i.e. the RF drive signal has eight independent components which are not combined but rather are connected independently to the first electrode by eight of the hollow conductors. Here the first and second fluid circulation channels may be jointly arranged to cool either side of the first arm, wherein the third and fourth fluid circulation channels may be jointly arranged to cool either side of the second arm, thereby to avoid a temperature gradient forming crossways between opposite sides of the first electrode.

(19) A convenient construction feature is for each hollow conductor to be provided with a tab located outside the vacuum housing via which it is supplied with the RF drive signal, e.g. by fixing one end of an inductive strap to the tab.
(20) The laser may further comprise a plurality of RF amplifiers and associated impedance matching circuits for generating a plurality of components of the RF drive signal, respective ones of the hollow conductors being connected to receive respective ones of the RF drive signal components output from respective ones of the impedance matching circuits. The RF amplifiers, impedance matching circuits, and hollow conductors may be connected to form one or more groups. Each group may consist of two RF amplifiers, two impedance matching circuits and two hollow conductors. Moreover, each said group can be associated with one circulation channel. Here one hollow conductor of a group supplies coolant fluid and the other extracts it. The laser may further comprise electrically insulating connector pieces connected to distal ends of the hollow conductors, the connector pieces being configured to connect with respective fluid supply lines.
(21) In further embodiments, the above-mentioned design features of the first electrode may be replicated for the second electrode. This may be of interest in particular for designs in which both the first and second electrodes are driven with RF, e.g. when the first and second electrodes are driven with respective RF drive signals that are out of phase with each other.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) In the following, the present invention will further be described by way of example only with reference to exemplary embodiments illustrated in the figures.
(2) FIG. **1**A is an exploded perspective view showing an RF slab laser according to an embodiment of the invention with an RF supply consisting of four RF amplifiers, an impedance matching network consisting of four impedance matching circuits, a four-way vacuum feed-through and a top electrode with two fluid circulation channels for cooling.
(3) FIGS. **1**B and **1**C are schematic plan and side views showing principal optical components of the RF slab laser of FIG. **1**A.
(4) FIG. **2** is a block diagram showing principal electrical components of the laser.
(5) FIG. **3** is a perspective view showing further details of the construction of FIG. **1**A.
(6) FIGS. **4**A and **4**B are plan and side section views of a hollow conductor.
(7) FIGS. **5**A to **5**D show a 4-way vacuum feed-through in plan view from above and below as well as perspective view from above and below.
(8) FIG. **6** is a circuit diagram of an impedance matching circuit.
(9) FIG. **7** is a circuit diagram of an impedance matching network consisting of four impedance matching circuits as shown in FIG. **6**.
(10) FIGS. **8**A to **8**D show an 8-way vacuum feed-through in plan view from above and below as well as perspective view from above and below.
(11) FIGS. **9** and **10** are schematic plan and perspective views of a top electrode and vacuum feed-through for a further embodiment in which the top electrode has four fluid circulation channels and is connected to an 8-way vacuum feed-through.
(12) FIGS. **11**A, **11**B and **11**C show three different implementation examples for arranging the hollow conductors to supply RF drive signal and coolant fluid to the top electrode.
DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS
(13) In the following detailed description, for purposes of explanation and not limitation, specific details are set forth in order to provide a better understanding of the present disclosure. It will be apparent to one skilled in the art that the present disclosure may be practiced in other embodiments that depart from these specific details.

(14) FIG. **1**A is an exploded perspective view showing an RF slab laser **100** according to an embodiment of the invention. FIGS. **1**B and **1**C are schematic plan and side views showing the principal optical components of the RF slab laser **100**. The laser **100** includes a vacuum housing **114** (shown partially cut away in FIG. **1**A) that forms a vacuum enclosure to contain a gaseous lasing medium. The lasing medium of a slab laser is a plasma of a gas, most commonly carbon dioxide as the active molecule, although other gases such as carbon monoxide, helium and nitrogen as well as gas mixtures containing one or more of these gases are known. The laser **100** further comprises a top electrode **102** driven with RF power and a bottom electrode **108** which is grounded, the two electrodes having respective inwardly facing surfaces **104**, **110** as well as outwardly facing surfaces **106**, **108** that face outwards towards the walls of the vacuum housing **114**. In the illustrated design, both electrodes **102**, **108** are arranged inside the vacuum housing **114**.
(15) The inwardly facing surfaces **104** and **110** of the top and bottom electrodes **102** and **108** are spaced apart by a gap of thickness ‘t’, and each have a width ‘w’ and a length ‘ls’ to form a slab (see FIGS. **1**B and **1**C). The gap is dimensioned to allow a plasma discharge **18**, shown schematically with the stippling, to be formed by applying an RF electrical drive signal to the top electrode **102** while the bottom electrode is grounded. In the following, the top electrode **102** is therefore sometimes referred to as the live, hot or driven electrode and the bottom electrode **108** as the ground(ed) electrode. A resonator cavity for the laser is formed by a pair of mirrors **20**, **22**. Mirror **20** is an end reflector of ideally 100% reflectivity. Mirror **22** is an output coupler which may provide output coupling by being partially transmissive (as would be a usual choice for a stable resonator design), or, as schematically illustrated, with edge coupling by configuring the cavity such that a small portion of the beam path passes by the lateral edge of the output mirror (as would be a usual choice for an unstable confocal resonator design) so that the output beam, labelled **24**, is emitted from the side of the output mirror **22**. Because of construction considerations, the cavity mirrors are placed outside the slab, so the cavity length is greater than the slab length. It will also be understood that the cavity mirrors may be planar or curved. A vacuum flange **70** for feed-through of the RF drive signal and the coolant fluid to the top electrode **102** extends into contact with the outwardly facing surface **106**.
(16) The vacuum flange **70** is sealingly mounted in an access aperture **116** in the vacuum housing **114**. The vacuum flange **70** has end portions of a plurality of hollow conductors **62** mounted therein. The top ends of the hollow conductors **62** are connected via electrically insulating pipe fitting connections **86**, e.g. made of plastic, to respective coolant pipes **88** which serve to carry coolant fluid, e.g. water, into and out of the vacuum housing **114**, where the water is circulated through or over the top electrode **102** via one or more fluid circulation channels **80**. Coolant fluid is thereby passed through the vacuum flange **70** around the top electrode **102** and then back out of the vacuum flange **70**. In FIG. **1**A, two such fluid circulation channels **80***a*, **80***b* are schematically shown. Each fluid circulation channel **80** is formed by internal passages **82** within the top electrode **102**. In the illustrated embodiment, there is one hollow conductor for input and one for output for each fluid circulation channel **80***a*, **80***b*. The internal passages **82** terminate in apertures **84** in the external (upper) surface **106** of the top electrode **102** and form fluid-flow connections with the proximal ends of the hollow conductors **62**. The vacuum flange **70** is arranged with its underside or bottom surface abutting or affixed to the outwardly facing surface of the top electrode **102**. The hollow conductors **62** that are fed through the vacuum flange **70** are arranged collectively so that they connect the top electrode **102** at or close to its geometric center (in plan view). This provides for symmetric dispersion of RF energy over the top electrode **102**. Alternatively, in other designs, the connections need not be positioned at the geometric center and could be anywhere along the outwardly facing surface of the top electrode **102**.
(17) Variants for the ground electrode **108** would include having the ground electrode **108** formed integrally with the vacuum housing **114** or with its outwardly facing surface **112** outside the vacuum enclosure, i.e. on the air side, and its inwardly facing surface **110** inside the vacuum

housing **114**. Variants for the live electrode **102** would include having the its outwardly facing surface **106**, or at least a portion thereof where the hollow conductors are connected, outside the vacuum enclosure and its inwardly facing surface **104** inside the vacuum housing **114**. The hollow conductors **62** need not all be connected to the outwardly facing surface **106** of the live electrode **102** in a confined area as in the present embodiment. Rather, in a variant in which at least a substantial proportion of the length of the outwardly facing surface **106** of the live electrode **102** is outside the vacuum, e.g. more than half or at least three quarters, the hollow conductors **62** may have their connections to the live electrode distributed, e.g. evenly spaced apart, along the exposed, i.e. air side, part of the length of the outwardly facing surface **106** of the live electrode **102**, e.g. singly or in pairs.

(18) FIG. **2** is a schematic block diagram showing the principal electrical components of the laser **100**. An RF source **5** is provided which comprises a plurality of RF power amplifiers **6**. The RF signals output from the RF power amplifiers **6** are supplied to associated impedance matching circuits **10** which together form an impedance matching network **8**. The outputs of the impedance matching circuits **10** are fed into the vacuum enclosure through the vacuum flange **70**. Each RF supply line is then connected separately and individually to the outwardly facing surface **106** of the top electrode **102**. The impedance matching network **8** has its circuit elements distributed between the PCB module **25**, the hollow conductors **62** and the vacuum flange **70**.

(19) FIG. **3** shows the PCB module **25** in more detail. Structurally, the PCB module **25** is built around a PCB **30**. The PCB module **25** has RF power connectors **12** which are the inputs into the impedance matching circuits **10** via which the outputs from the RF power amplifiers **6** are received. The PCB **30** has an at least approximately centrally positioned aperture **34**, i.e. a through-hole, which is dimensioned and positioned to allow the hollow conductors **62** to pass therethrough. The PCB module **25** accommodates components for four impedance matching circuits **10**, these being arranged in four quadrants of the PCB **30** around the aperture **34**. Each RF power connector **12** feeds the RF drive current received from its associated RF power amplifier **6** through an associated inductive strap (or ribbon or coil) **56** having a first end **52** and a second end **54**, the inductive strap **56** forming a first inductor element. The first ends **52** are electrically connected to the upper surface of the PCB **30**, for example on one of the conductive pads **32**. The second ends **54** are located at or adjacent the PCB aperture **34** and are connected to respective ones of the hollow conductors **62** (not shown in FIG. **3**). The PCB **30** is additionally used as a basis for forming one or more capacitors **45** for each of the impedance matching circuits **10**. Namely, the PCB **30** forms a dielectric for the capacitors and the plates of each capacitor are formed respectively on the top and bottom sides of the PCB **30** by top conductive pads **32** and one or more bottom conductive pads **40**. One of the conductive pads **32** of each impedance matching circuit **10** is connected to its RF power connector **12**, thereby connecting each conductive pad **32** to its associated RF power amplifier **6**. The bottom conductive pad **40** can be a single contiguous plate shared commonly by all the capacitors. Alternatively, a separate conductive pad **40** could be formed for each of the conductive pads **32** or for groups thereof. The inductive straps **56** are each configured to be easily removable so that different ones can be exchanged to trim the impedance matching when setting up the laser. The available inductive straps **56** may differ in shape, length and/or some other parameter which changes their inductance to allow an appropriate selection to be found.

(20) FIGS. **4**A and **4**B are plan and side section views of a single one of the hollow conductors **62**. The hollow conductor **62** may be conveniently formed by a section of copper pipe to which is soldered or otherwise affixed a connection tab **67**, which allows a second end **54** of one of the inductive straps **56** to be fastened thereto. The fastening may be done with fastening bolts (not shown) which may be screwed into threaded holes in the lateral tabs **67**.

(21) FIGS. **5**A-**5**D show a vacuum flange **70** in plan view from above and below as well as perspective view from above and below. The vacuum flange **70** comprises a base **74**, which has the form of a circular disc and is a solid piece made of metal or a metallic material, a sleeve **75**, which

has the form of a hollow cylinder enclosing the hollow conductors **62**, and an electrically insulating connector **76**, which has the form of a band or ring and is made, for example, of a ceramic material. The vacuum flange **70** is four-way, i.e. has four hollow conductors **62** integrated into it. The vacuum flange sleeve **75** is made of metal in our examples manufactured to date, but could alternatively be made of a ceramic material or other electrically insulating material as well as a metallic material. The ring-shaped connector **76** forms a junction piece between the base **74** and sleeve **75**, and is inserted in between them. The upper and lower end surfaces of the ring-shaped connector may be metallized to facilitate respective solder joints being formed with the base **74** and with the sleeve **75**. The outer surface of the ring-shaped connector **76** forms a vacuum-tight connection with the access aperture **116** in the vacuum housing **114**. The hollow conductors **62**, which are arranged extending parallel to each other, extend out of the flange **70** and are located in a vacuum-tight manner within the flange base **74**, i.e. a length portion of each hollow conductor **62** is embedded in the flange base **74**, e.g. with a suitable solder connection. The bottom ends of the hollow conductors **62**, i.e. the vacuum-side ends, may terminate flush with the inside surface of the flange base **74**, i.e. on its vacuum side, or extend out a certain distance from the inside surface of the flange base **74** (as shown in FIG. **5**D). It is noted that the lateral tabs **67** are located on the air-side of the vacuum flange **70**. The hollow conductors **62** and the flange base **74** collectively provide a second inductor through their self-inductance.

(22) FIG. **6** shows a circuit diagram of a single one of the impedance matching circuits **10**. A capacitor **45** of capacitance C**1** is implemented by one or more conductive pads **32** and the common conductive pad **40** with the PCB **30** acting as the dielectric. A first inductor **50** of inductance L**1** is provided by one of the inductive straps **56**, the inductance of which is adjustable as necessary be swapping out different ones from the available set of inductive straps **56**. A second inductor **60** of inductance L**2** is provided by the hollow conductor **62** and its mounting in the vacuum flange **70**. The second inductance L**2** is fixed in the present design and depends on the outer diameter and the length of the hollow conductors **62**. It is therefore possible to design in a desired fixed value for L**2** by adjusting these dimensions. Increasing the outer diameter of the hollow conductors **62** reduces the inductance L**2** and increasing the length of the hollow conductors **62** increases the inductance L**2**. The first and second inductors **50** and **60** are electrically connected to the RF power amplifier **6** in series and the capacitor **45** is electrically connected to the RF power amplifier **6** in parallel with the inductors **50** and **60**. Optionally, the impedance matching circuit **10** may additionally include one or more reactances or reactive impedance components (resistive, capacitive, inductive or a combination of these) (not shown), which may be arranged on the PCB **30**. The reactances can be electrically connected to any of: the RF power connectors **12**, the conductive pads **32**, the inductive straps **56** and the hollow conductors **62** either in parallel or in series, as necessary for achieving desired impedance matching between the RF power amplifier **6** and the top electrode **102**.

(23) FIG. **7** is a circuit diagram showing the impedance matching network **8** consisting of four of the impedance matching circuits **10** shown in FIG. **6** which deliver four RF supply lines through the vacuum flange **70** as described above. Capacitors C**1***a*, C**1***b*, Clc and C**1***d* are the capacitors implemented by conductive pads **32** and **40**. First inductors L**1***a*, L**1***b*, L**1***c* and L**1***d* are respective ones of the first inductors **50** formed by the inductive straps **56**, whose inductance is changeable as needed. Second inductors L**2***a*, L**2***b*, L**2***c* and L**2***d* are respective ones of the second inductors **60** formed by respective ones of the hollow conductors **62**. The capacitors C**1***a*, C**1***b*, C**1***c* and C**1***d* are electrically connected to four RF power amplifiers **6** in parallel. The first inductors L**1***a*, L**1***b*, L**1***c* and L**1***d* and the second inductors L**2***a*, L**2***b*, L**2***c* and L**2***d* are electrically connected to the four RF power amplifiers **6** in series. As described in U.S. Pat. No. 9,263,849B2, this configuration of an RF source **5** with multiple RF power amplifiers **6** and associated impedance matching circuits **10**, which are each connected individually and separately to adjacent locations on the top electrode **102**, serves to deliver multiple RF power supply lines at multiple locations on the top electrode **102**. This can significantly reduce RF power loss compared with a standard design where multiple

RF power supplies have their outputs combined outside the vacuum enclosure and fed through a single impedance matching circuit before being delivered to the top electrode through a single supply line. The heat generated by the impedance matching circuit is proportional to the square of the current. As a consequence a design having ‘n’ independent RF supply lines with associated impedance matching circuits that supply to ‘n’ locations on the top electrode means that each supply line carries one n-th of the total current injected into the live electrode, instead of a single RF supply line output from a single impedance matching circuit. This reduces the amount of heat generated by a factor ‘n’, e.g. a factor of four in the example of having four pairs of RF power amplifier and impedance matching circuit.

(24) FIGS. **8**A-**8**D correspond to FIGS. **5**A-**5**D and show another example vacuum flange **70** which differs from that of FIGS. **5**A-**5**D in that there are eight hollow conductors instead of four.

(25) FIGS. **9** and **10** show in plan and perspective view a top electrode **102** and vacuum flange **70**, the vacuum feed-through being as shown in FIGS. **8**A-**8**D. The top electrode **102** is provided with internal passages that form four fluid circulation channels **80***a*, **80***b*, **80***c*, **80***d* each with its own input and output apertures **84** on the external (top) surface of the electrode **102**. A useful feature of this design with four fluid circulation channels is that the cooling is relatively even across the electrode both lengthways and crossways when viewed in plan view and also symmetric both lengthways (x-direction) and crossways (y-direction), i.e. symmetric about both a centrally positioned crossways plane Y0 and a centrally positioned lengthways plane X0 as illustrated. For example, to provide a more uniform temperature difference between the fluid circulation channels, the flow paths can be arranged such that the outward flow paths are arranged laterally on the inside and the inward flow (return) paths are arranged laterally on the outside. The coolant fluid entering the electrode, which is cold, is therefore piped through the more central part of the electrode **102** body (closer to the X0 plane), where the electrode would be the hottest absent any cooling, whereas the warmer coolant fluid returning from the electrode ends flows closer to the sides of the electrode (farther away from the X0 plane), thereby reducing the temperature gradient crossways across the electrode. Although there may be some temperature gradient lengthways along the electrode as the distance increases from the central location of the vacuum flange **70** (i.e. distance from Y0 plane) as a consequence of the coolant fluid warming up, any such lengthwise temperature gradient will be gradual.

(26) FIGS. **11**A, **11**B and **11**C are schematic section views showing three different implementation examples for arranging the hollow conductors to supply RF drive signal and coolant fluid to the top electrode.

(27) FIG. **11**A shows an arrangement as described above and shown, for example in FIGS. **5**A-**5**D. The vacuum housing **114** encloses a vacuum space which is indicated by stippling in the drawing. The top electrode **102** is arranged inside the vacuum space and by way of example two of the internal passages **82** are shown, these forming the channels for circulating coolant fluid around the top electrode **102**. The passages **82** terminate in apertures in the outwardly facing surface of the top electrode. A ring-shaped connector **76** forms a butting, vacuum-tight connection with the vacuum flange base **74**. A vacuum flange sleeve **75** is arranged in butting, vacuum-tight connection over the ring-shaped connector **76**. An upper portion of the outer surface of the vacuum flange sleeve **75** forms a vacuum-tight connection with the inside rim surface of the vacuum housing aperture **116** via an O-ring **78** which sits in an annular groove **79** formed in the top surface of the vacuum housing **114** adjacent its aperture **116**. The O-ring **78** is squashed down and laterally extended to form the vacuum-tight connection between vacuum housing **114** and sleeve **75** via a flange piece **77** that is screwed down on the top surface of the vacuum housing **114** as indicated schematically by the downward arrows. Length portions of the hollow conductors **62** are embedded in the vacuum flange base **74** in a vacuum-tight manner. The vacuum flange base **74** is made of electrically conducting material. The inwardly facing surface of vacuum flange base **74** is connected with the outwardly facing surface of the top electrode **102** in a fluid-tight and vacuum-

tight manner. The proximal ends of the hollow conductors **62** align with the fluid input and output apertures of the passages **82** to provide a fluid-tight connection for the coolant fluid circulation paths. With this implementation, since the bottom, inwardly facing surface of the vacuum flange base **74** abuts or is fixedly connected to the outwardly facing surface of the top electrode **102** in a fluid-tight and vacuum-tight manner, all that is needed for the coolant fluid circulation path to be formed is alignment of the passage apertures and the proximal ends of the hollow conductors and a fluid-tight connection where they meet. The proximal ends of the hollow conductors also need to be in good electrical contact with the top electrode in order for the RF drive signal to be transmitted.

(28) FIG. **11**B shows an alternative implementation example which is described by comparison with the example of FIG. **11**A. In this example, the electrically insulating connector **76** extends from the access aperture **116** of the vacuum housing **114** down into the vacuum space all the way to the outwardly facing surface of the top electrode **102** where it forms a vacuum-tight seal with the first electrode. The electrically insulating connector **76** is shaped and dimensioned in plan view to match the shape and dimensions of the access aperture **116**, so in the illustrated example has the shape of a cylindrical tube to match the circular shape of the access aperture **116**. A suitable ceramic material may be used for the connector **76**. The vacuum-tight connection between the electrically insulating connector **76** and the top electrode **102** may be effected by any suitable means. For example, at its proximal end the ceramic tube connector **76** could be vacuum-sealed to the top electrode by pushing it into an annular groove cut into the top of the electrode which accommodates a sealing O-ring (not shown). Another option would be to braze or otherwise affix an intermediate metal ring to the proximal end of the ceramic tube connector **76** to allow the intermediate metal ring to be brazed to the top electrode **102**. An advantage of this implementation is that the hollow conductors do not enter the vacuum space, so their proximal end connections to the top electrode **102** are also outside the vacuum space, so only need to be fluid-tight, i.e. they do not need to be vacuum-tight. This may be effected with any suitable means; by way of example the drawing shows O-rings seated in counterbored recesses at the input and output apertures. It is further noted that in the illustration a base **74** is included which has the hollow conductors embedded into it but which is not functionally part of the vacuum flange **70**. The vacuum flange **70** in this embodiment consists solely of the electrically insulating connector **76**. The base **74** in the illustrated example merely serves to mechanically and electrically connect the hollow conductors, which may be convenient but is not essential. In variants of this design, the illustrated base **74** may be omitted.

(29) FIG. **11**C shows a further alternative implementation example which is described by comparison with the examples of FIGS. **11**A and **11**B. The vacuum seal to the vacuum housing **114** is implemented in a similar manner to FIG. **11**A with a flange piece **77** that is screwed down on the top surface of the vacuum housing **114** by clamping bolts **71** which pass through the flange piece **77** and screw into threads in the top surface of the vacuum housing **114** thereby to press and distend the O-ring **78** to form a vacuum-tight connection between the inside rim surface of the vacuum housing aperture **116** and the outside surface of the vacuum flange sleeve **75**. The O-ring **78** sits in an annular groove **79** formed in the top surface of the vacuum housing **114** adjacent its aperture **116**. A suitable alternative to using an O-ring would be to use an indium gasket or wire. The vacuum-tight seal to the outwardly facing surface **106** of the top electrode **102** is however formed differently compared to the design of FIG. **11**A. The vacuum flange sleeve **75** has an outer flange at its base, i.e. proximal the top electrode **102**, which has through holes to allow the vacuum flange sleeve **75** to be bolted down on the top electrode **102** with clamping bolts **72**. One part of the electrical isolation between the (metal) vacuum flange sleeve **75** (which is grounded) and the top electrode (which is live) is ensured through the provision of upper and lower annular ceramic spacers **73** through which the clamping bolts **72** pass. The clamping bolts (which are live) are thus electrically isolated from the vacuum flange sleeve **75**. The lower ceramic spacers **73** have a

cylindrical form. The upper ceramic spacers are cup-like to support the underside of the bolt heads of the bolts **72**. Another part of the electrical isolation between the (metal) vacuum flange sleeve **75** (which is grounded) and the top electrode (which is live) is ensured through the provision of an annular electrically insulating connector **76** with the shape of a hollow cylinder which is sandwiched between the base of the vacuum flange sleeve **75** and the outwardly facing surface **106** of the top electrode **102**. A suitable material for the insulating connector **76** is a ceramic material. As illustrated a locating groove for the insulating connector **76** may be formed in the outwardly facing surface **106** of the top electrode **102** as well as a similar locating groove in the base of the vacuum flange sleeve **75**. A vacuum seal is formed between the upper surface of the insulating connector **76** and the lower surface of the base of the vacuum flange sleeve **75** and a further vacuum seal is formed between the lower surface of the insulating connector **76** and the outwardly facing surface **106** of the top electrode **102**. There are various ways to form the two metal-ceramic vacuum seals including: brazing, soldering, providing an indium gasket or wire, and providing an O-ring. Moreover, in this example, the hollow conductors **62** are directly embedded in the uppermost depth portion of the top electrode **102** and penetrate to form a fluid flow connection with respective ones of the internal coolant passages **82** for liquid supply or removal. The end portions of the hollow conductors **62** that are embedded in the top electrode **102** form fluid-tight connections. Similar to the example of FIG. **11**B, an advantage of this implementation is that the hollow conductors do not enter the vacuum space, so their proximal end connections to the top electrode **102** are also outside the vacuum space, so only need to be fluid-tight, i.e. they do not need to be vacuum-tight.
(30) Various modifications from the specific embodiments described above may be envisaged.
(31) An embodiment can be envisaged with one fluid circulation channel supplied by two hollow conductors electrically connected to two impedance matching circuits supplied from two RF power amplifiers. Generally a convenient arrangement is when one pair of hollow conductors is associated with one fluid circulation channel. Moreover, the number of circulation channels that pass coolant around the top electrode can be freely chosen, i.e. is not restricted to 2 or 4 as in the illustrated examples, but may be any number from 1 to 5, 6, 7, 8 or more. Furthermore, alternative designs for the coolant passages in the top electrode may be envisaged in which a single hollow conductor has its fluid flow split into two (or more) as it enters the top electrode to supply two (or more) different coolant channels within the top electrode. These two (or more) coolant channels could remain separate downstream and have their fluid extracted by two (or more) hollow conductors. Alternatively, these two (or more) coolant channels could come back together downstream where they terminate at an aperture in the outwardly facing surface of the top electrode and have their coolant fluid extracted by a single hollow conductor. In other words, designs can be envisaged which do not have a two-to-one ratio of the number of hollow conductors to fluid circulation channel(s).
(32) Moreover, it will be understood that the same arrangement as described in detail above for the top electrode may be replicated for the bottom electrode. This may be of interest in particular for designs in which both the top and bottom electrodes are driven with RF, i.e. when an RF source is connected to the outwardly facing surfaces of both the top and bottom electrodes, the bottom electrode thereby being supplied with an additional RF electrical drive signal that is out of phase with the RF electrical drive signal supplied to the top electrode.
(33) It will be clear to one skilled in the art that many improvements and modifications can be made to the foregoing exemplary embodiments without departing from the scope of the present disclosure.
REFERENCE NUMERALS
(34) **5** RF supply **6** RF power amplifier **8** impedance matching network **10** impedance matching circuit **12** RF power connector **18** plasma discharge **20** rear mirror **22** front mirror (output coupler) **24** output beam **25** PCB module **30** printed circuit board (PCB) **32** top conductive pads **34** aperture

40 bottom conductive pad(s) **45** capacitor **50** first inductor **52** first end of **56** **54** second end of **56** **56** inductive strap (or ribbon or coil) **60** second inductor **62** hollow conductor **67** connection tab of hollow conductor **66** **70** vacuum flange **71** clamping bolts for clamping flange **77** **72** clamping bolts for outer flange of vacuum flange sleeve **75** **73** annular ceramic spacers for clamping bolts **72** **74** vacuum flange base (e.g. circular disc) **75** vacuum flange sleeve (e.g. hollow cylinder with optional integral outer flange) **76** electrically insulating connector (e.g. ceramic ring) **77** clamping flange **78** O-ring **79** groove in outer surface of vacuum housing **80** fluid circulation channel (**80***a*, **80***b*; **80***a*, **80***b*, **80***c*, **80***d*) **82** internal passages in cooled electrode **84** apertures in outwardly facing surface of cooled electrode **86** pipe fitting union connections (plastic) **88** coolant pipes C**1** capacitor L**1** first inductor L**2** second inductor **100** RF slab laser **102** top electrode **104** top electrode inwardly facing surface **106** top electrode outwardly facing surface **108** bottom electrode **110** bottom electrode inwardly facing surface **112** bottom electrode outwardly facing surface **114** vacuum housing **116** access aperture

Claims

1. A radio frequency, RF, slab laser comprising: a first electrode and a second electrode having respective inwardly and outwardly facing surfaces, wherein their respective inwardly facing surfaces face each other and are spaced apart by a gap forming a slab waveguide of a thickness dimensioned to allow a plasma discharge to be formed by driving at least the first electrode with an RF drive signal, the first electrode being provided with at least one fluid circulation channel for distributing coolant fluid around the first electrode; a vacuum housing having an access aperture adjacent the outwardly facing surface of the first electrode and enclosing at least the inwardly facing surfaces of the first and second electrodes inside a vacuum space; a plurality of hollow conductors connected to the outwardly facing surface of the first electrode to supply the RF drive signal to the first electrode and the coolant fluid to the at least one fluid circulation channel of the first electrode; and a vacuum flange arranged in the access aperture to form a vacuum-tight seal with the vacuum housing and a further vacuum-tight seal with the outwardly facing surface of the first electrode.
2. The laser of claim 1, wherein the vacuum flange comprises a sleeve, an electrically insulating connector and a base, the base having an intermediate length portion of each hollow conductor embedded therein in a vacuum-tight manner, and the electrically insulating connector being connected with respective vacuum-tight connections between the sleeve and the base, wherein the sleeve forms said vacuum-tight seal with the vacuum housing and wherein the base forms said further vacuum-tight seal with the outwardly facing surface of the first electrode.
3. The laser of claim 1, wherein the vacuum flange comprises an electrically insulating connector extending from the access aperture, where it forms said vacuum-tight seal with the vacuum housing, to the outwardly facing surface of the first electrode, where it forms said further vacuum-tight seal with the outwardly facing surface of the first electrode, thereby defining a portion of the outwardly facing surface of the first electrode that is outside the vacuum space to which the hollow conductors are connected.
4. The laser of claim 1, wherein the vacuum flange comprises a sleeve and an electrically insulating connector, wherein the sleeve forms said vacuum-tight seal with the vacuum housing and wherein the insulating connector forms said further vacuum-tight seal with the outwardly facing surface of the first electrode.
5. The laser of claim 1, wherein each fluid circulation channel comprises at least one internal passage formed within the first electrode having ends that terminate in fluid input and output apertures in a surface of the first electrode, the fluid input and output apertures being arranged in fluid-flow connection with proximal ends of the hollow conductors.
6. The laser of claim 5, wherein the fluid input and output apertures are arranged aligned with the

proximal ends of the hollow conductors.
7. The laser of claim 1, wherein the at least one fluid circulation channel comprises further hollow conductors arranged in thermal contact with the outwardly facing surface of the first electrode and in fluid-flow connection with the hollow conductors.
8. The laser of claim 1, wherein the first electrode has an elongate shape with a length at least ten times greater than its width and the hollow conductors connect to the first electrode at least approximately mid-way along the outwardly facing surface of the first electrode, thereby subdividing the first electrode into first and second arms.
9. The laser of claim 8, wherein the at least one fluid circulation channel comprises a first fluid circulation channel extending around the first arm of the first electrode and a second fluid circulation channel extending around the second arm of the first electrode.
10. The laser of claim 9, wherein the first fluid circulation channel is connected to be supplied with coolant fluid by first and second ones of the hollow conductors and wherein the second fluid circulation channel is connected to be supplied with coolant fluid by third and fourth ones of the hollow conductors.
11. The laser of claim 8, wherein the at least one fluid circulation channel comprises first and second fluid circulation channels extending around the first arm of the first electrode and third and fourth fluid circulation channels extending around the second arm of the first electrode.
12. The laser of claim 11, wherein the first fluid circulation channel is connected to be supplied with coolant fluid by first and second ones of the hollow conductors and wherein the second fluid circulation channel is connected to be supplied with coolant fluid by third and fourth ones of the hollow conductors, wherein the third fluid circulation channel is connected to be supplied with coolant fluid by fifth and sixth ones of the hollow conductors and wherein the fourth fluid circulation channel is connected to be supplied with coolant fluid by seventh and eighth ones of the hollow conductors.
13. The laser of claim 11, wherein the first and second fluid circulation channels are jointly arranged to cool either side of the first arm, wherein the third and fourth fluid circulation channels are jointly arranged to cool either side of the second arm, thereby to avoid a temperature gradient forming crossways between opposite sides of the first electrode.
14. The laser of claim 1, further comprising a plurality of RF amplifiers and associated impedance matching circuits for generating a plurality of components of the RF drive signal, respective ones of the hollow conductors being connected to receive respective ones of the RF drive signal components output from respective ones of the impedance matching circuits.
15. The laser of claim 14, wherein the RF amplifiers, impedance matching circuits, and hollow conductors are connected to form one or more groups, each group consisting of two RF amplifiers, two impedance matching circuits and two hollow conductors.
16. The laser of claim 15, wherein each said group is associated with one circulation channel.